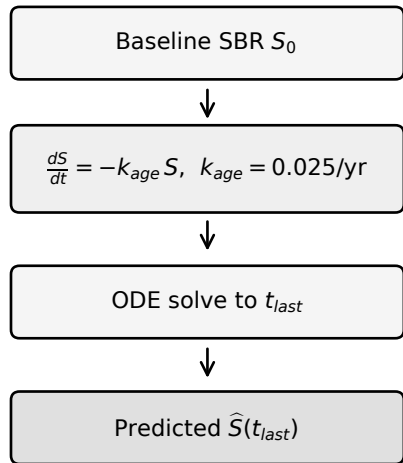
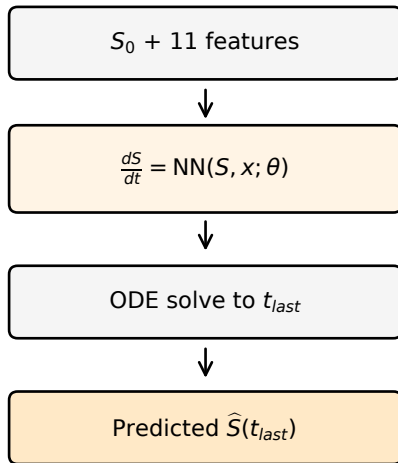
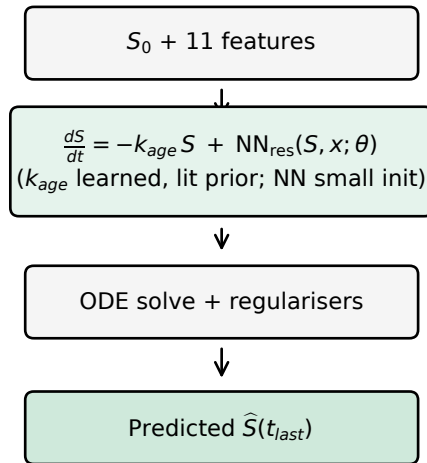


(a) Pure mechanistic

test MAE 0.205

(b) Pure Neural ODE

test MAE 0.157

(c) Physics-informed hybrid (UDE)

$$\mathcal{L} = \text{MSE} + \lambda_{\text{phys}} \|\text{NN}_{\text{res}}\|^2 + \lambda_{\text{mono}} \text{ReLU}(dS/dt)$$

best: $\lambda_{\text{phys}} = 0.1$, $\lambda_{\text{mono}} = 0.1$

test MAE 0.152